Kaiwen Bian

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Education

2022 – 2026 **University of California, San Diego**, La Jolla, CA
B.S. in Data Science, Major-GPA 3.891/4.0
B.S. in Cognitive Behavioral Neuroscience, Major-GPA 3.875/4.0
Research Interests: Embodied Agent, Neural Foundation Model, Computational Neuroscience, Geometric Learning.

Research Experience

2026 – Now **Stanford University**, *Full-time Researcher*, Palo Alto, CA.
Enigma Project, Department of Ophthalmology & Stanford Bio-X
Advisor: Andreas S Tolias
- Building brain foundation models.

2024 – 2026 **Salk Institute for Biological Studies**, *Research Intern*, La Jolla, CA.
Crick-Jacobs Center for Theoretical and Computational Biology
Advisor: Talmo D Pereira
- Developed computationally efficient deep reinforcement imitation systems (Mimic-MJX) to control a bio-mechanically realistic agent and mimic realistic animal behaviors.
- Discovered governing topological insights into the representational space of embodied agents' behaviors (TopoMimic) and achieved higher performance on behavior transition clustering compared to SOTA segmentation methods.
- Built a switching nonlinear dynamical system (Code2Act) that maps discrete behavioral syllables to continuous control through learned internal dynamics.

2025 – 2026 **Stanford University**, *Visiting AI Researcher*, Palo Alto, CA.
Statistics Department & Wu Tsai Neurosciences Institute
Advisor: Scott W Linderman
- Created latent dynamical models for planning and generating bio-mechanically realistic behaviors for embodied agents using deep state-space modeling methods.
- Co-advised on the Code2Act project from Salk.

2025 – 2026 **University of California, San Diego**, *Student Researcher*, La Jolla, CA.
Halıcıoğlu Data Science Institute
Advisor: Yusu Wang, Gal Mishne
- Created the Graph Compositional Abstraction Scheme for Tokenization (MOSAIC), allowing sequence transformers to generate complex molecules more realistically with more structural properties compared to SOTA flat tokenization methods.
- Co-advised on the TopoMimic project from Salk.

2023 – 2024 **University of California, San Diego**, *FMP Scholar*, La Jolla, CA.
Cognitive Science Department & Undergraduate Research Hub
Advisor: Sean Trott
- Developed tools using linguistic techniques (i.e. affordance) for probing the embodied simulation in large multi-modal models to improve the interpretability of these models.

Conference & Journal Publications

ICLR 2027 **Bian, K.**, Zhang, C. Y., Yang, Y., Sirbu, A., Jha, A., Buchanan, K., Leonardis, E. J., Richards, B. A., Ölveczky, B. P., Linderman, S. W., & Pereira, T. D. (2026). Code to Action: Motor Programs from Discrete Syllables in Whole-Body Biomechanical Control. *Manuscript in preparation to submit to International Conference on Learning Representations.*

Nature 2026 Zhang, C.Y., Yang, Y., Sirbu, A., Leonardis, E.J., Abe, E., Warnberg, E., Aldarondo, D.E., Lee, A., Prasad, A., Foat, J., **Bian, K.**, Park, J., Bhatt, R., Saunders, H., Nagamori, A., Thanawalla, A.R., Huang, K.W., Plum, F., Beck, H., Flavell, S.W., Labonte, D., Richards, B.A., Brunton, B.W., Azim, E., Ölveczky, B.P., & Pereira, T.D. (2026). MIMIC-MJX: Neuromechanical Emulation of Animal Behavior. Manuscript submitted to *Nature Methods*.

NeurIPS 2026 **Bian, K.**, Yang, A. H., Parviz, Ali., Mishne, G., & Wang, Y. (2026). Beyond Flat Walks: Compositional Abstraction for Autoregressive Graph Generation. *Manuscript submitted to Advances in Neural Information Processing Systems.*

TAG-DS 2026 **Bian, K.**, Leonardis, E. J., Yang, Y., Zhang, C., Azim, E., Ölveczky, B. P., Wang, Y., & Pereira, T. D. (2026). Shaped by What's Missing: Topological Invariance Simplification Discovers Behavioral Transitions. *Manuscript in preparation to submit to Conference of Topology, Algebra, and Geometry in Data Science.*

Conference Presentations

SimBodies 2026 **Bian, K.**, Zhang, C. Y., Yang, Y., Sirbu, A., Jha, A., Buchanan, K., Leonardis, E. J., Richards, B. A., Ölveczky, B. P., Linderman, S. W., & Pereira, T. D. (2026). Code to Action: Motor Programs from Discrete Syllables in Whole-Body Biomechanical Control. *Accepted at Simulated Bodies: Whole Body Biomechanical Models (SimBodies) Conference, Lansdowne, VA.*

UCSD 2026 **Bian, K.**, Yang, A. H., Parviz, Ali., Mishne, G., & Wang, Y. (2026). Beyond Flat Walks: Compositional Abstraction for Autoregressive Molecular Generation. *Accepted at Halıcıoğlu Data Science Institute, UC San Diego Senior Capstone Showcase, La Jolla, CA.*

COSYNE 2026	Zhang, C., Sirbu, A., Yang, Y., Leonardis, E. J., Park, J., Prasad, A., **Bian, K.**, Abe, E., Wärnberg, E., Brunton, B. W., Richards, B., Ölveczky, B. P., & Pereira, T. D. (2026). MIMIC-MJX: Neuromechanical Emulation of Animal Behavior. *Poster will be presented at Computational and Systems Neuroscience (Cosyne) Conference, Lisbon, Portugal.*
COSYNE 2026	Leonardis, E. J., Nagamori, A., Thanawalla, A., Yang, Y., Saunders, H., Gilmer, J., Zhang, C., **Bian, K.**, Ölveczky, B. P., Al Borno, M., Azim, E., & Pereira, T. D. (2026). Musculoskeletal Imitation Learning: Physics-Aware Constraints Promote Naturalistic Muscle Activity. *Accepted at Computational and Systems Neuroscience (Cosyne) Conference, Lisbon, Portugal.*
Stanford 2025	**Bian, K.**, Jha, A., Buchanan, K., & Linderman, S. W. (2025). Deep State Space Controls For Biomechanically Realistic Artificial Agents. *Poster presented at Stanford Undergraduate Research Program (SURP-Stats) Symposium, Palo Alto, CA.*
SfN 2025	**Bian, K.**, Leonardis, E. J., Yang, Y., Zhang, C., Azim, E., Ölveczky, B. P., Wang, Y., & Pereira, T. D. (2025). Topology-driven Insights into Naturalistic Behavior from Neuromechanical Agent Modeling. *Poster presented at the Society for Neuroscience (SfN) Annual Meeting, San Diego, CA.*
SfN 2025	Yang, Y., Zhang, C., Leonardis, E. J., Sirbu, A., **Bian, K.**, Azim, E., Ölveczky, B. P., & Pereira, T. D. (2025). VNL-playground: Fast and Biologically Realistic Virtual Environment for Simulating Animal Behavior. *Poster presented at the Society for Neuroscience (SfN) Annual Meeting, San Diego, CA.*
SfN 2025	Leonardis, E. J., Nagamori, A., Thanawalla, A., Yang, Y., Saunders, H., Gilmer, J., Zhang, C., **Bian, K.**, Ölveczky, B. P., Al Borno, M., Azim, E., & Pereira, T. D. (2026). Musculoskeletal Imitation Learning: Physics-Aware Constraints Promote Naturalistic Muscle Activity. *Poster presented at the Society for Neuroscience (SfN) Annual Meeting, San Diego, CA.*
COSYNE 2025	Zhang, C., Yang, S., **Bian, K.**, Abe, E., Wärnberg, E., Foat, J., Aldarondo, D., Brunton, B. W., Ölveczky, B. P., & Pereira, T. D. (2025). Track-MJX: A GPU-Accelerated Pipeline for Imitating Animal Motor Control. *Poster presented at the Computational and Systems Neuroscience (COSYNE) Conference, Montreal, Canada.*
UCSD 2023	**Bian, K.**, Li, A., Jones, C., & Trott, S. (2023). Embodied Simulation in Multimodal Models Using Affordance Stimulus: A Probing Study. *Poster presented at the UC San Diego Undergraduate Research Faculty Mentorship Program (FMP) Symposium, La Jolla, CA.*

Skills

Programing	Python, SQL, Java, Java Script, Web Dev.
Python Packages	Jax, Flax, Ray, MuJoCo/MJX, Gym, Dm-Control, Brax, PyTorch (Torch, TorchRL, and PyG), TensorFlow, Ripser, SkLearn, Panda, Dask, and Spark.
Mathematic	Linear Algebra, **Machine/Deep Learning**, **Reinforcement Learning**, **Geometric Learning** (Graph Learning, Computational Topology, and Manifold Learning), Probability/Statistics (State Space Models, Probabilistic Inference, Stochastic Processes, and Computational Stochastic), Convex Optimization, Dynamical System, Signal Processing, and Linear Control Theory.
Neuroscience & Biology	System & Cognitive Neuroscience, Neuro-anatomy, Motivational & Developmental Neurobiology, Behavioral Endocrinology, Neural Signal Processing, Exercise Physiology, Kinesiology, and Human Nutrition.
Language	Mandarin Chinese and English